

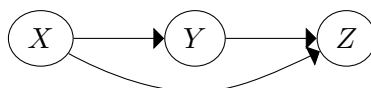
Topic 16: Graphical Representations of Random Variables

02-680: Essentials of Mathematics and Statistics

July 30, 2026

1 Bayesian Networks

Sometimes we need a more complicated conditional description. One way to do this is using a network (graph).



In general for a network such as this we define the probabilities as:

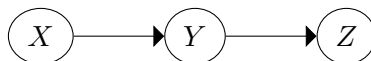
$$p(\langle Q_1, Q_2, \dots, Q_d \rangle) = \prod_{i=1}^d p\left(Q_i \mid \bigwedge_{j \in N^{in}(Q_i)} Q_j\right)$$

remember that $N^{in}(v)$ is the set of in-neighbors of v in a graph.

So for the Bayesian Network above, we may have something like:

$p(X = x)$		$p(Y \mid X)$		$p(Z \mid Y, X)$				
x		x	$y = 1$	$y = 0$	x	y	$z = 1$	$z = 0$
0	0.7	0	0.5	0.5	0	0	0.3	0.7
1	0.3	1	0.9	0.1	0	1	0.1	0.9
					1	0	0.7	0.3
					1	1	0.4	0.6

But if we alter the network to the following (i.e. remove the dependence of Z on X),



we end up with

x	$p(X = x)$	$p(Y X)$			$p(Z Y)$		
x		$y = 1$	$y = 0$		$z = 1$	$z = 0$	
0	0.7	0	0.5	0	0.2	0.8	
1	0.3	1	0.9	1	0.7	0.3	

2 Markov Chains

Markov chains are usually used to model sequences of items. We saw when talking about conditional probabilities that:

$$p(X_n, X_{n-1}, \dots, X_1) = p(X_n | X_{n-1}, \dots, X_1)p(X_{n-1}, \dots, X_1) \quad (1)$$

$$= p(X_n | X_{n-1}, \dots, X_1)p(X_{n-1} | X_{n-2}, \dots, X_1)p(X_{n-2}, \dots, X_1) \quad (2)$$

$$\dots \quad (3)$$

$$= p(X_n | X_{n-1}, \dots, X_1) \dots p(X_2 | X_1)p(X_1) \quad (4)$$

$$(5)$$

The Markov assumption is that the probability of X_k depends on X_{k-1} alone. This simplifies the computation, but at the cost of long term connections. Anecdotally the Markov assumption says:

the future is independent of the past given the present.

So that means we can rewrite the statement above as:

$$p(X_n, X_{n-1}, \dots, X_1) = p(X_n | X_{n-1})p(X_{n-1} | X_{n-2}) \dots p(X_2 | X_1)p(X_1).$$

This is the basis for Hidden Markov Models which are used a lot in genetics.

Example. Consider a sequence of DNA nucleotides modeled as a set of random variables $N_1 N_2 \dots N_n$, each of which can take on the value $\{A, T, C, G\}$. We want to know if the sequence is a CpG island or not.

Looking at the equation above, if we know the initial distribution (that is the probability distribution for $p(N_1)$) for both cases, and the transition distributions ($p(N_k | N_{k-1})$) we can determine which is more probable.

$$P_b = \begin{bmatrix} & A & C & G & T \\ A & 0.30 & 0.20 & 0.29 & 0.21 \\ C & 0.32 & 0.30 & 0.08 & 0.30 \\ G & 0.25 & 0.25 & 0.29 & 0.21 \\ T & 0.18 & 0.24 & 0.29 & 0.29 \end{bmatrix} \quad P_c = \begin{bmatrix} & A & C & G & T \\ A & 0.18 & 0.27 & 0.43 & 0.12 \\ C & 0.17 & 0.37 & 0.27 & 0.19 \\ G & 0.16 & 0.34 & 0.37 & 0.13 \\ T & 0.08 & 0.36 & 0.38 & 0.18 \end{bmatrix}$$

Assuming for both $p(N_1 = \mathbf{A}) = p(N_1 = \mathbf{C}) = p(N_1 = \mathbf{T}) = p(N_1 = \mathbf{G}) = 0.25$, what is the sequence **ACTTC** more likely to be a CpG island or not?

Useful References

Degroot and Schervish. “Probability and Statistics” §§3.10